Assignment 3

1. Create a LinkedList and insert elements at the end.
   Testcase:
   Existing LinkedList: [5, 10, 15]
   Elements to insert: [20, 25]
   Expected Output: LinkedList: 5 → 10 → 15 → 20 → 25

2. Insert a new node at the beginning of a LinkedList.
   Testcase:
   Existing LinkedList: [10, 20, 30]
   Node to insert: 5
   Expected Output: LinkedList: 5 → 10 → 20 → 30

3. Insert a new node at a given position in a LinkedList.
   Testcase:
   Existing LinkedList: [10, 20, 30, 40]
   Node to insert: 25 at position 2
   Expected Output: LinkedList: 10 → 20 → 25 → 30 → 40

4. Delete the first node of a LinkedList
   Testcase:
   Existing LinkedList: [10, 20, 30, 40]
   Expected Output: LinkedList: 20 → 30 → 40

5. Delete the last node of a LinkedList.
   Testcase:
   Existing LinkedList: [10, 20, 30, 40]
   Expected Output: LinkedList: 10 → 20 → 30

6. Delete a node by its value in a LinkedList.
   Testcase:
   Existing LinkedList: [10, 20, 30, 40]
   Node to delete: 30
   Expected Output: LinkedList: 10 → 20 → 40

7. Search for an element in a LinkedList.

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Testcase:
Existing LinkedList: [10, 20, 30, 40]
Element to search: 30
Expected Output: Element found at index 2

8. Count the total number of nodes in a LinkedList.

Testcase:
Existing LinkedList: [10, 20, 30, 40]
Expected Output: Total nodes: 4

9. Reverse a LinkedList.
Testcase:
Existing LinkedList: [10, 20, 30, 40]
Expected Output: LinkedList: 40 → 30 → 20 → 10

Existing LinkedList: []
Expected Output: LinkedList: (empty)

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